

Business Continuity Plan - Settlement Operations

Reference BCP-2026-M. Owner: Payments Operations. Effective 2026-05-01. This plan covers continuity of the Vireo Clearing Hub, the Kestrel settlement service, and the Halcyon ledger. Product-level commercial commitments live in PRD-2027-A; this plan defines what happens when they are at risk.

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1 Incident severity scale

All settlement-affecting incidents are classified on a four-level scale at detection and reclassified as facts emerge. Severity drives paging, communication cadence, and post-incident obligations.

Severity	Definition	Paging	Member communication
SEV-1	Hub-wide settlement stall or data-integrity doubt in Halcyon	Auto-pages Payments Operations on-call; no manual step	Within 15 minutes, then every 30 minutes
SEV-2	Degraded settlement (finality misses targets) or one window at risk	On-call paged by duty officer	Within 60 minutes, then hourly
SEV-3	Single-participant impact or redundant-component loss	Business hours follow-up	Affected participant only
SEV-4	Cosmetic or documentation defect	Ticket queue	None

The Vireo Duty Officer may raise severity unilaterally but may lower it only with the incident commander's agreement. A KSN-ERR-503 storm affecting more than five participants is SEV-1 by definition, per the error semantics in KSN-2027-A section 4.

2 Recovery objectives

Recovery objectives are set per service, not per system. The funding-account ledger tolerates zero data loss: settlement finality is only ever declared after a durable Halcyon write, which is why the RPO below is zero.

Service	RTO	RPO	Notes
Kestrel settlement (prime window)	15 minutes	0	Failover target to standby site
Kestrel settlement (off-peak)	60 minutes	0	Batch reconciliation may shift within the window
Halcyon ledger and reconciliation	30 minutes	0	Ledger is the recovery source of truth
Aviary portal	4 hours	24 hours	Portal outage never blocks settlement
Osprey list refresh (SCR-2026-A)	24 hours	1 refresh cycle	Stale-list operation beyond 24h escalates to E4

3 Site strategy

The Vireo Clearing Hub runs active-standby across two geographically separated data centers, designated VCH-A (primary) and VCH-B (standby). Physical locations are withheld from general circulation under the site-security standard and are available to Internal Audit on request. Halcyon replicates synchronously to VCH-B; settlement finality is acknowledged only after both sites confirm the write.

3.1 Failover decision

- Automatic failover: triggered by loss of VCH-A heartbeat for 60 seconds during any settlement window.
- Manual failover: incident commander plus Vireo Duty Officer jointly authorize; single-person failover is prohibited.
- Failback to VCH-A happens only outside settlement windows and never on a quarter-end freeze day.

3.2 Degraded-mode rules

While failed over to VCH-B, Kestrel accepts instructions normally but the single-instruction ceiling for Tier-K3 participants is temporarily halved to 375,000 KES-eq to bound sponsor risk, and the off-peak finality target relaxes from 30 to 60 seconds. Participants receive the standard degradation notice referencing KSN-ERR-503 semantics: no manual resubmission, auto-replay applies.

4 Roles

Role	Held by	Authority
Incident commander	Senior Payments Operations engineer on rota	Owens severity, failover, and the incident timeline
Vireo Duty Officer	24/7 rota, reachable via the Kestrel operations channel	First responder; may raise severity unilaterally
Communications lead	Network Participation duty manager	Owens member notices and the status page
Scribe	Assigned at incident start	Maintains the timeline in Halcyon incident records

5 Communication templates

Member notices follow fixed templates so integrators can parse them: NTC-1 (degradation), NTC-2 (failover executed), NTC-3 (all-clear with reconciliation summary). Every notice names the affected windows in EAT and states explicitly whether the freeze-period rules of KSN-2027-A section 3 are altered. During SEV-1, the 15-minute first notice is a hard commitment even when facts are incomplete.

6 Testing and drills

- Full site-failover drill: quarterly, in the off-peak window of the second month, alternating automatic and manual triggers.
- Ledger-recovery restore test: monthly, restoring the previous day's Halcyon snapshot to an isolated environment and re-running reconciliation.
- On-call paging test: weekly, Monday 09:00 EAT.
- Participant-facing drill: annually, coordinated through Aviary, exercising the NTC notice chain end to end.

A failed quarterly failover drill is itself a SEV-3 and blocks the next production change window until the failure is dispositioned.

7 Post-incident review

Every SEV-1 and SEV-2 gets a written post-incident review within five business days: timeline from the scribe's Halcyon record, root cause, participant impact including any Halcyon clawbacks raised, and corrective actions with owners. Reviews are shared with the EAPC for SEV-1 incidents as a regulatory obligation, and internally summarized in the monthly KSN-STAT pack.

8 Upstream dependencies

Settlement continuity depends on services Meridian does not fully control. Each dependency has a documented degraded-mode behavior; none is allowed to be a silent single point of failure.

Dependency	Used for	Degraded-mode behavior
EAPC regulatory gateway	Designation feeds, statutory reporting	Osprey refresh runs on last-received feed; reporting queues locally
Inter-site fiber pair (two providers)	VCH-A to VCH-B replication	Single-provider loss: no impact; dual loss triggers automatic failover evaluation
Hardware security module fleet	Instruction signing, token issuance	Quorum of 3 of 5 HSMS required; below quorum settlement halts as SEV-1
Time source (dual GNSS + holdover clock)	Finality timestamps, drill validation	Holdover keeps drift under 250 ms for 72 hours
Primary telco SMS gateway	On-call paging	Secondary provider auto-engaged; weekly Monday test covers both

9 Failover runbook summary

The full runbook lives in the operations console; this summary exists so that severity calls and member communications can proceed if the console itself is unreachable. Steps are strictly ordered.

Step	Action	Owner	Target time
F1	Confirm VCH-A loss (heartbeat, out-of-band check)	Vireo Duty Officer	T+2 min
F2	Declare SEV-1, page incident commander	Vireo Duty Officer	T+3 min
F3	Authorize failover (automatic trigger or joint manual call)	Incident commander + Duty Officer	T+5 min
F4	Promote VCH-B; verify Halcyon replication watermark is current	On-call engineer	T+10 min
F5	Resume settlement in degraded mode (section 3.2 limits)	Incident commander	T+15 min (the Kestrel prime-window RTO)

Step	Action	Owner	Target time
F6	Issue NTC-2 member notice	Communications lead	T+15 min

Failback is not part of this runbook: it is a planned change executed outside settlement windows under standard change control, per section 3.1.

Appendix A - Drill scenario library

Quarterly failover drills rotate through this library so that no scenario goes untested for more than two years. Scenario S3 is mandatory in any quarter following a real SEV-1.

Scenario	Simulates	Last run	Result
S1	Clean VCH-A power loss during prime window	2027-02	Green
S2	Creeping degradation: rising latency without heartbeat loss	2026-11	Amber - detection took 11 minutes, above the 5-minute target
S3	Failover with in-flight quarter-end freeze	2026-08	Green
S4	HSM quorum loss below 3 of 5	2027-05	Green
S5	Dual fiber loss with GNSS outage (compound)	2026-05	Amber - holdover clock validated, member notice late at 22 minutes

Amber results generate corrective actions tracked to closure in the following quarter's drill report; two consecutive ambers on the same scenario escalate to the Chief Risk Officer.

Revision history

Version	Date	Change
1.0	2024-09-15	Initial plan, single-site with cold standby
2.0	2026-05-01	Current edition: active-standby VCH-A/VCH-B, synchronous Halcyon replication, degraded-mode Tier-K3 limits
2.1	2027-06-20	Scenario S5 added after the 2026-05 compound drill